

Things (IoT), and sustainability. All these areas demand a range of semiconductor innovations, from high-performance GPUs to chiplets and 3D packaging. Technology and business trends drive each other, fuelling the semiconductor industry's rapid expansion.

SG: How do you view the geopolitical landscape in semiconductors, especially with recent supply chain disruptions and self-reliance efforts by many countries post-Covid?

SS: The supply chain issue is a result of the USA-China tension. Countries are now focused on building domestic capabilities to reduce dependency, leading to fragmented manufacturing and supply chain models. The US has invested \$52 billion through the CHIPS Act, while Europe and other nations are implementing similar policies. This localisation effort is primarily driven by strained US-China relations, not just Covid-related disruptions. If geopolitical concerns with China were resolved, the global



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semiconductor supply chain could remain collaborative.

SG: What impact will China's pivot to mid-range manufacturing have on mid-tier semiconductor companies like Microchip?

SS: China has responded to its limited access to advanced manufacturing technology by focusing on mid-range capabilities, building significant capacity for trailing-edge technologies. We are seeing many Chinese startups entering the market, especially in commodity products, which could bring some competition. However, about half of Microchip's business in China involves Western firms manufacturing there, which are now shifting operations to countries like Vietnam and Thailand. Around 5% of our China business is at risk of local competition, but Microchip's diverse portfolio and quality-driven approach should mitigate significant impact.

SG: How do you see emerging technologies, like chiplets and 3D packaging, affecting the cost-sensitive sub-\$10 chip market?

SS: Chiplets and 3D packaging are advantageous but expensive, so they are currently more viable for high-end applications, such as data centres and AI processors. As adoption grows, prices may drop, gradually making these technologies accessible to mid-range markets. However, they are unlikely to penetrate low-cost microcontrollers and analogue chips because the price reduction would still be insufficient to support such low-margin products.

SG: RISC-V has gained popularity, especially among enthusiasts and educational institutions, but has yet to make a major commercial impact. How do you see it evolving?

SS: RISC-V has gained momentum, with many major companies now working on RISC-V projects. A significant milestone was when NASA selected RISC-V architecture for its next-generation space processors. While RISC-V was once an alternative among many architectures, it is now positioned to compete with ARM, primarily due to ARM's licensing fees and other restrictions. RISC-V offers an open-source model, which we expect to gain substantial traction in the coming years. Microchip has integrated RISC-V selectively in

